

Magic You

Tongze Guo
Department of Digital Media, Beijing
Normal University
China
tongze@mail.bnu.edu.cn

Wen Zhou
Department of Digital Media, Beijing
Normal University
China
zhouwen@bnu.edu.cn

Xiangrong Xiao
Art and Technology Innovation
Center, Beijing Normal University
China
xiangrong73@vip.sina.com

Ni Ding
Department of Digital Media, Beijing
Normal University
China
dingni@bnu.edu.cn

Yansong Chen
Department of Digital Media, Beijing
Normal University
China
cys598394662@gmail.com

Shaolong Liu
Department of Digital Media, Beijing
Normal University
China
lsl@mail.bnu.edu.cn



Figure 1: Screenshots of real-time scenes

Abstract

"Magic You" is a virtual reality interactive narrative experience, a coming-of-age personal story about ADHD, with joys, anxieties, bitter memories, and growing pains, told in a hand-drawn, colored-pencil aesthetic as a magical journey with surreal overtones, providing a more in-depth look into the inner world of people with ADHD. The artwork hopes to present the experiences and imagination of ADHD patients to the audience in poetic and romantic way, make audience spiritual healing and empathizing.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality; • **Applied computing** → Arts and humanities; Media arts; • **Social and professional topics** → User characteristics; People with disabilities.

Keywords

Virtual Reality, Interactive narratives, ADHD, Immersive storytelling

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1 Introduction

The artwork tells the story of a person with ADHD from the age of 1 year old to adulthood, from growing up and discovering that he has unusual magic, to becoming confused about himself, to enduring the pressure and words of others, and ultimately using this magic to reconcile with it. It includes four main scenes of growing up, full of magic and surreal imagery, from a traditional Chinese ritual scene to a middle school classroom, office, and a surreal giant tower. The scenes and objects designed for the artwork come from research and interviews with ADHD patients, and the interactions in which the paper balls are turned into paper planes come from the patient's childhood memories, and the interaction and artistic scenes effects design according to the two main symptoms of ADHD: "Attention-deficit" and "hyperactivity", each scene corresponds to a certain extent to certain conditions in different stages. At the same time, this experience is also visually distinctive, as well as interactively. The artwork uses hand-tracking to not only complete ordinary grabbing, but also combines the paper plane-launching of interaction with climbing, exploring, and developing the vertical spatial dimension of VR, adding a lot of narrative clues to the vertical space, and the paper planes can take the viewer to a lot of

places outside of the main narrative line, encouraging the viewers to explore freely.

2 PROJECT INNOVATIONS

2.1 Interactive Design

We introduced the important interaction effect of "Paper Plane - Launch", so that the audience can only use Pinch and grip gestures to complete a shooting aiming function, which has a stronger sense of participation and experience in the work. At the same time, the climbing function was developed with hand-tracking, in most of the previous VR -works, VR climbing is basically as a unique function of the handle in the game field, but we will climb this way using hand-tracking to express, and to a certain extent, the use of hand-tracking on the procedure of the optimization of the gripping coordinates to avoid the hand-tracking range of the climb caused by the failure.



Figure 2: Screenshots of some interactions in the artwork

2.2 Vertical Spatial Design

We used vertical space to carry the narrative, which is rare in previous XR experiences. By using paper planes and climbing, we expanded the audience's possibilities of exploring vertical space, and had the potential to solve VR vertigo.

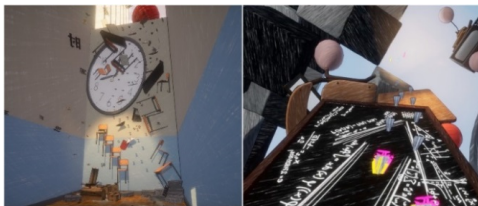


Figure 3: Vertical spatial experience in the artwork

2.3 Artistic Expression of ADHD

The symptoms of ADHD are mainly manifested as Attention Deficit Disorder(ADD) and Hyperactivity Disorder(HD) [Furman, 2005], which are given some magical artistic effects, for example, when the player gazes at the objects in the scene of "Zhua Zhou", the peripheral objects will appear as flicking artful lines; the books that are unreadable; and the scattered falling words and so on [Hartmann et al., 2016]. ADHD is reflected in the restless objects

that often appear in the scene, and the special effects that appear when the player is delivering paper planes. In this artwork, we have hidden a lot of details to imply the viewer's own identity, and artistically represent some symptoms of ADHD, and we also want to use this to show the world of ADHD, which is not a world to be pitied by others, but also a beautiful world in their eyes, which reflects neurological diversity, not a reflection of defects.



Figure 4: Artistic expressions in the artwork

3 TEST AND OPTIMIZATION

During the production process, several internal tests and questionnaire (SPES) [Hoogman et al., 2020] were carried out to optimize the presence of the work as well as interactive narrative aspects. A total of 11 right-handed dominant subjects (8 females and 3 males, mean age 22) participated in the test.

Table 1: VR Interactive Experience (SPES)

VR interactive experience	Mean±SD
Interactive system	3.73±0.27
Interactive UI	4.08±0.47
Interactive function	3.64±0.57
Interactive content	3.93±0.57
Spatial Presence	3.84±0.64
Emotional involvement	3.22±0.33

(1 = completely disagree, 2 = disagree, unsure = 3, agree = 4, strongly agree = 5), with closer to 5 meaning that a more positive experience was gained in the certain aspect gained.

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References

- Lydia Furman. 2005. What Is Attention-Deficit Hyperactivity Disorder (ADHD)? *Journal of Child Neurology* 20, 12 (December 2005), 994–1002. <https://doi.org/10.1177/08830738050200121301>
- Tilo Hartmann, Werner Wirth, Holger Schramm, Christoph Klimmt, Peter Vorderer, André Gysbers, Saskia Böcking, Niklas Ravaja, Jari Laarni, Timo Saari, Feliz Gouveia, and Ana Maria Sacau. 2016. The Spatial Presence Experience Scale (SPES). *Journal of Media Psychology* 28, 1 (January 2016), 1–15. <https://doi.org/10.1027/1864-1105/a000137>
- Martine Hoogman, Marije Stolte, Matthijs Baas, and Evelyn Kroesbergen. 2020. Creativity and ADHD: A review of behavioral studies, the effect of psychostimulants and neural underpinnings. *Neuroscience & Biobehavioral Reviews* 119, (December 2020), 66–85. <https://doi.org/10.1016/j.neubiorev.2020.09.029>